

The effect of gap on the quality of glass-to-glass welding using a picosecond laser

Jiejuan Zhang^{a,*}, Shanghang Chen^b, Hanlun Lu^c, Minghe Huang^a, Jiaming Li^a, Liang Guo^a, Qitao Lue^b, Qingmao Zhang^{a,*}

^a Key Laboratory of Nanophotonic Functional Materials and Devices of Guangdong Province, South China Normal University, Guangzhou 510006, Guangdong, China

^b Han's Laser Technology Industry Group Co., Ltd. Shenzhen 518057, Guangdong, China

^c Guangdong Institute of Semiconductor Industrial Technology, Guangzhou 510651, Guangdong, China

ARTICLE INFO

Keywords:

Microwelding

Ultrafast laser

Glass

Air gap

ABSTRACT

In order to study the influence of air gap and focus position on the quality of ps-laser glass-to-glass welding we implemented a method of detecting the air gap between glass surfaces based on optical thin film interference. The method allows real-time optimization of the welding process. We measure the air gap between glass surfaces to be bonded and the laser focal position, which are set as the constraint conditions affecting the nonlinear self-focusing of the laser beam. The processing parameters of picosecond laser welding are determined and the welding strength and the morphology of bonded zone are detected and observed. Laser welding with good performance can be achieved with an air gap of 2 μm . SEM microscopy shows that the bonding region between two glass surfaces is very smooth in the vicinity of the tear-drop shaped cross-sectional structure of the welded glass. A tensile strength inspection fixture is designed, and we measured a tensile fracture strength of over 30 MPa for the welded glass. Through confocal microscope analysis, the width and depth of fracture morphology of welded samples are shown to change with air gap, which is consistent with the theoretical analysis.

1. Introduction

Optically transparent substrates such as glass, sapphire, diamond, etc. play an indispensable role in daily life and scientific research. With excellent structural strength, resistance to wear, exceptional corrosion resistance and outstanding optical and electrical compatibilities, these transparent materials have become the substrates of choice for MEMS components, bio-chips, TFT panels and capacitive touch pads. Bonding of homogeneous or heterogeneous transparent substrates is one of the key procedures in manufacturing the devices mentioned above. Traditional bonding techniques including anodic bonding, brazing and use of adhesives have significant drawbacks like creep, out-gassing and aging [1–4]. Ultrafast laser welding can overcome these problems with the additional advantages of highly localized heating [5,6], flexible and easy processing. In particular, ultrafast laser welding can realize direct microwelding without the need to insert an intermediate layer. The microwelding exploits the nonlinear absorption of laser radiation due to high peak intensities [5,6], whilst avoiding excessive heating of surrounding material due to the highly localized melt zone. As a new packaging technology, ultrafast laser welding receives significant attention for MEMS, electronic components, biochips and capacitance sensors.

In the welding of amorphous (glass) materials, there are many factors such as laser power and focal position which affect weld quality. However the size of the gap between the two glass surfaces is an important factor restricting industrial application of this welding technology. Many studies of glass welding since the work of Tamaki et al. [7] in 2005 employed optical contact [8,9] between the glass surfaces. Early studies on ultrafast laser welding had identified a requirement for the gap to be in the range of 100 to 500 nm [10–17] to avoid cracking or ablation at the interface [18]. In order to eliminate the harsh conditions of optical contact, in 2015, Jianyong Chen et al. [19] successfully welded borosilicate glass and fused silica respectively with a pre-existing gap of 3 μm using a sample with a curved surface (cylindrical lens) together with a glass wafer, or two wafers, one which has a series of grooves etched into the surface. Chen et al. showed that the maximum gap that can be welded is strongly dependent on the laser power and focal position and they also proposed a mechanism to explain the observations. In 2011, Kristian Cvecek et al. present optical contact as a prebonding method for glass welding yielding higher seam strength [20]. The area not in optical contact can be easily distinguished from the optical contact area, as the interference fringes can be observed on the surface of glasses with gap distances in a range of a few micrometers within the coherence length of thermal light. The interference fringes can be used to calculate the gap distance. In 2015, Kristian Cvecek et al. proposed

* Corresponding authors at: School of Information and Optoelectronic Science and Engineering, South China Normal University, Guangzhou 510006, China.

E-mail address: zhangqm@scnu.edu.cn (Q. Zhang).

a method utilizing ultra-short laser pulses which exploits a volume expansion of irradiated glass enabling the joining of glass plates across gaps of up to $1\text{ }\mu\text{m}$ [21]. This joining procedure establishes Van-der-Waal forces as well as chemical bonds between the joining partners, thus exceeding the pure Van-der-Waals bonding energy of a pure optical contact. The gap width can be approximately determined by analyzing the number and sequences of the color interference fringes of natural light source. The prepared gap heights for all samples were between 250 nm and $1\text{ }\mu\text{m}$. In 2019, Marion Gstalter et al. proposed a long focal length focusing device with the associated Rayleigh length method for glass welding by femtosecond laser pulses [22]. 30 MPa tensile strength of glass-glass welding, for long focal length (no microscope objective), up to $3\text{ }\mu\text{m}$ gap, with speed up to 1000 mm/s have been reported.

Crack formation prevention was introduced in a review by Kristian Cvecek in 2019 [23]. If a gap exists during welding, residual tensile stresses will be present in the welded specimen. Glass are usually about 20 times more resilient against compressive stresses than against tensile stresses. In the preliminary research work, initial efforts were made effort to reduce the gap by outside mechanical pressure or optical contacting. However, with the development of laser technology, the scanning system with long focal depth can realize the welding of glass with large gaps and no accompanying cracks. Therefore, it is very important to detect the solderable distance between glasses to explore the physical mechanism of the interaction between ultrafast laser and transparent material. With the expansion of gap, the physical and chemical changes in the melting process will be affected, as well as the plasma generation process. As soon as the plasma region is enclosed by a melt pool, no further cracks will form, let alone propagate outside the melt pool [24].

The advent of three-dimensional glass structures, together with widespread application of flexible glass in electronic components, make it difficult to control the air gap between glass surfaces, a fact that will directly affect weldability and the performance of glass welds. Therefore, it is necessary to measure the air gap between glass surfaces, so as to predict and evaluate the welding performance in engineering applications. Furthermore, challenging processes such as the welding of multilayer glass and improving the reliability of electronic components by using flexible glass have increased the need for better theoretical analysis of welding processes. Research into nonlinear laser transmission and absorption in the microscopic weld region will be helpful for the design of fixtures, optimization of process parameters, achievement of high weldability, and realization of the automation of the ultra-fast

laser welding process. At present, for the welding process with certain air gap (non-optical contact area), the influence of slight change of gap on the microstructure and performance of the welding has not been reported.

In this study, we report a method to detect the thickness of the air layer between glass surfaces, based on the principle of interference. The method allows real-time detection of the gap between surfaces to be welded, which is helpful for the realization of process automation. We have concentrated on understanding the impact on the self-focusing phenomenon of laser radiation by an air gap and on studying the propagation of ultrafast laser beams, enabling us to control the absorption rate and welding performance. This research will greatly expand the real-world applications of ultrafast laser welding.

2. Experimental Setup

The experiment was performed using a mode-locked Yb:YAG laser (IPG) of 1064 nm wavelength, 2 ps pulsewidth and average output power of 50 W , focused with a stationary lens ($f=24\text{ mm}$) to micro-weld glass to glass. Compared with previous work [25], the experimental processing system has undergone some improvements. The laser beam energy density has been increased by reducing the beam diameter from $20\text{ }\mu\text{m}$ to $7\text{ }\mu\text{m}$. The actual used power of laser in the presented experiment is 5 W . This beam is focused onto a movable processing platform with a movement accuracy of $2\text{ }\mu\text{m}$ along the X and Y axes and $5\text{ }\mu\text{m}$ along the Z axis. The travel along the X, Y and Z axes is 300 , 250 and 20 mm , respectively, and the range of moving speed is $1\text{--}500\text{ mm/s}$ in the X and Y directions and $0.1\text{--}2.5\text{ mm/s}$ in the Z direction of the loading platform. The scanning speed of galvanometer scanners is 1 mm/s – $10,000\text{ mm/s}$. There is a device to control acceleration and deceleration in the galvanometer scanners system. In the experiment, we first turned on the laser for a period of time to stabilize the scanning speed, and then welded the glass plates to ensure stable processing speed.

The focal position is also determined by a CCD vision system. The dimensions of each substrate were $30\times 10\text{ mm}^2$, while the thicknesses of Corning third generation gorilla glass was 0.50 mm .

The preliminary treatment of the samples is described in detail in paper [25]. The series of interference fringes, which appear when the surfaces of the samples are not in optical contact, can indicate the thickness of the air layer between the two opposing glass surfaces. This highly accurate welding system can achieve satisfactory welding across the thin air gap between two glass surfaces. We provide a detailed test for the thickness of an air layer between two glass samples (Fig. 1).

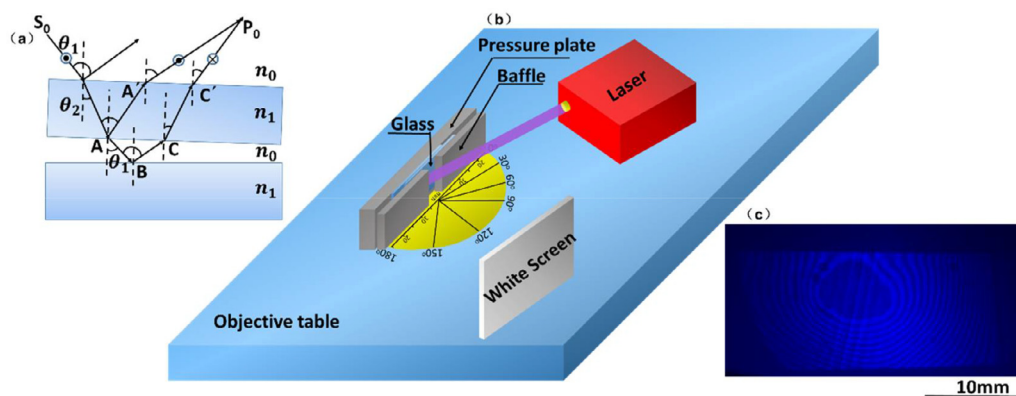


Fig. 1. (a) Schematic of light transmission (b) Schematic of air gap measurement (c) Interference fringes.

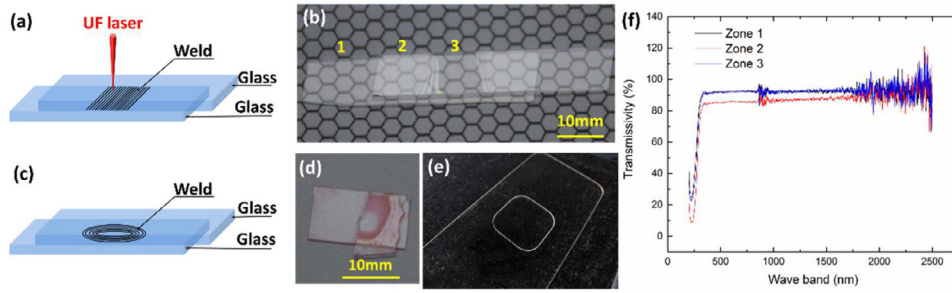


Fig. 2. Samples of glass welding by ultrafast laser (a) Parallel lines welding (b) Welded sample. (c) Circular welding (d) Welded samples after heating and beating (e) Welded sample of a phone screen (f) Transmissivity of different zones in (c).

According to the interference principle, we designed a setup to measure the air thickness between opposing surfaces of two glass samples, as shown in Fig. 1(b). The laser has strong coherence, monochromaticity and high directivity. Compared with white light, it is easier to produce interference fringes with obvious contrast, which is convenient to calculate the air gap according to the sequence of interference fringes. Therefore, we chose laser at 405 nm as the illumination source. A CW laser at 405 nm with power of 400 mW is incident at a chosen angle onto the interface between two glass samples, which are clamped together by a pressure plate and baffle in a darkened environment. The size of the laser-irradiated area is controlled by adjusting the position of the baffle. The interference fringes are viewed on the white screen (Fig. 1c). The thickness of the air gap can be calculated by counting fringes.

The paths of the rays that form the interference fringes are shown in Fig. 1(a). A laser beam is sent from S_0 , and the two beams reflected by the interface surface of two glasses intersect at P_0 , generating interference fringes. The optical path difference is:

$$\Delta = n_0(AB + BC + C'P - A'P) + n_1(CC' - AA') \quad (1)$$

Because the air gap is usually very small and the plates are very nearly parallel, we can use the parallel plate calculation formula approximately instead. Because the two intersecting beams of light have phase shift, the optical path difference is

$$\Delta = 2n_0h \cos \theta_1 + \frac{\lambda}{2} \quad (2)$$

For a given incidence angle, the optical path difference only depends on the thickness (h) of plate at the reflected area. Therefore, the interference fringe spacing depends on the thickness of the air layer. The bright

fringe positions satisfy:

$$\Delta = 2n_0h \cos \theta_1 + \frac{\lambda}{2} = m\lambda \quad (m = 1, 2, \dots) \quad (3)$$

and the dark fringe positions satisfy:

$$\Delta = 2n_0h \cos \theta_1 + \frac{\lambda}{2} = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, 1, 2, \dots) \quad (4)$$

By substituting fringe positions into formulae (3) and (4), the air gap height at the corresponding position can be calculated in the weldable area on the sample. In addition to the conventional testing equipment for welding samples presented in paper [25], we designed a tension chuck with linear guide to accurately measure the tensile properties of welded samples, which will be introduced in detail later in this study. We also tested the transparency of the welded area and used a simple method suitable for industrial application to test the sealing performance of welded samples.

3. Results and discussion

One of the biggest applications of glass welding by ultrafast lasers is packaging of mobile phone screens. Fig. 2(e) is a photo of a phone screen onto which we welded a small piece of glass. The transparency of the welded area between two glass plates is very good.

The transparency of the glass in a device display is very important because it affects the clarity of the display. Two pieces of glass were joined together successfully as shown in Fig. 2(a). The welding sample has length of 30 mm and width of 10 mm. The background pattern (4 mm pitch) is visible through the sample (Fig. 2(b)). In Fig. 2(f), zone1 is the optical transmission through the original glass substrate, zone 2

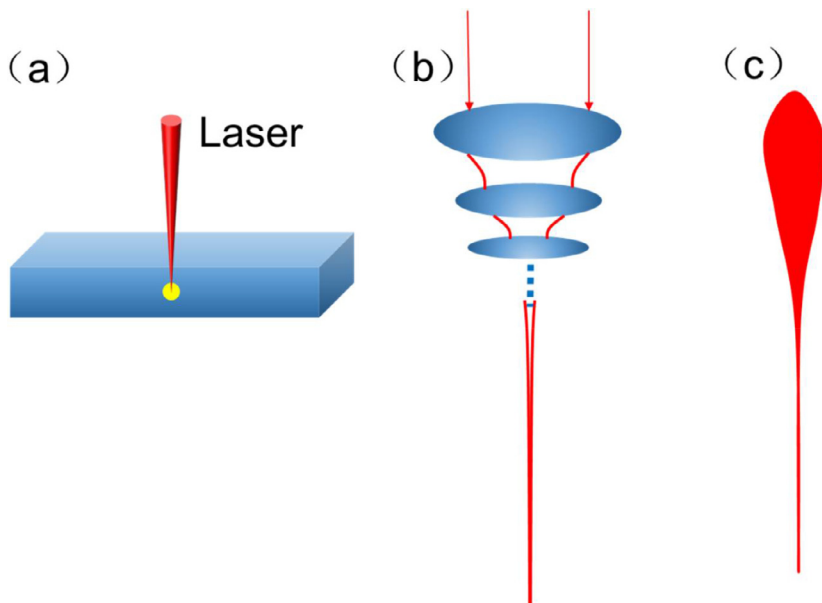


Fig. 3. Self-focusing phenomenon of an ultrafast laser pulse in glass (a) One piece of glass is illuminated by ultrafast laser (b) A real photorefractive effect (c) A filamentous structure.

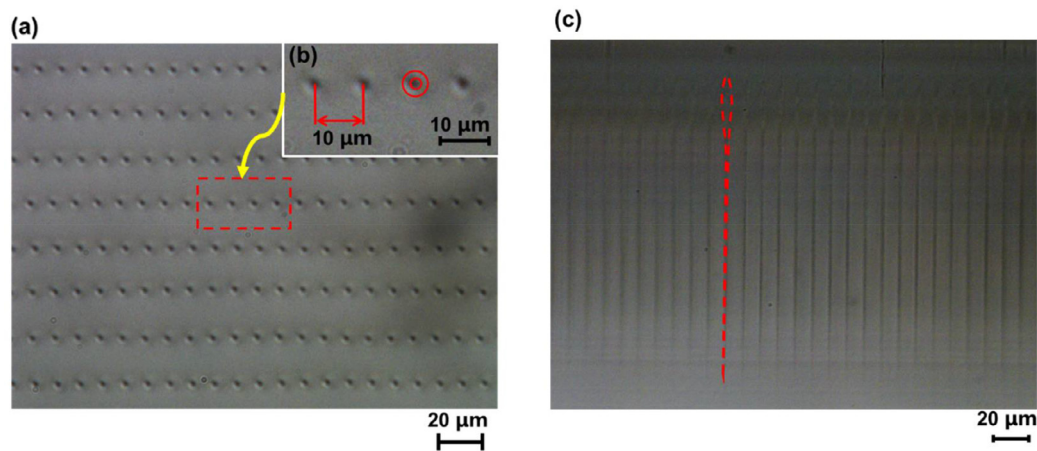


Fig. 4. Morphology of radiation region by laser pulses inside glass (a) and (b) Top view; (c) Side view.

is the transmission through the two glass samples before welding, and zone 3 is the transmission through the welded area after welding. By comparing the clarity of the image of the honeycomb structure through three zones, we found that the transparency of the laser-welded zone was almost identical to that of the original single glass substrate. There is some loss due to reflection in zone 2 due to the influence of the air layer between two glasses. Because of the good bonding between two glasses which eliminates the air layer, there is no additional reflection in zone 3. As seen in Fig. 2(f), the light transmittance across the visible and near infrared is up to 97% in zones 1 and 3 compared with 90% in zone 2.

The sealing performance of the welded area is also very important whether in mobile phone screens or electronic packages. Two glass samples were welded together using welds arranged in a ring structure to test the sealing performance (Fig. 2(c)). The sample before doing this simple hermetic test is similar to that shown in Fig. 2(b) except that the welding zone 3 has a circular welding structure as shown in Fig. 2(c). We tested the sealing performance of the seal by placing the welded sample in a solution with a ratio of carbon red ink to water of 1:1. After heating and boiling for 2 h, we dropped the welded sample from a height of 1 m onto a hard surface. The processed sample is shown in the Fig. 2(d). There are traces of red liquid coloring on the periphery of the weld, almost no red liquid in the welded area, and the transparency of the non-welding area in the middle of the ring is also very good. The sealing performance of welding area is good from macroscopic point of view. A more sophisticated test of seal performance will be described in

a future publication. Because the ultrafast laser deposits energy locally, the temperature in most of the assembly is unchanged, which allows temperature-sensitive materials or components such as polymers, metals, or electronic payloads to be encased without damage.

In order to further explore the mechanism of glass welding by ultrafast laser, and so as to guide the process and promote the industrial application of this technology, we created a new model. We consider the welding of two pieces of glass as a process in which the ultra-fast laser is focused on a slab of glass into which a thin layer of air with a refractive index of 1 has been inserted. In this new model, we explore how the thickness and position of this air layer (i.e. the distance between the opposing surfaces of the glass and the focal position of laser) affects the interaction between the ultrafast laser pulses and glass, and thus the welding performance. Using this model, we first investigated the effect of ultrafast laser pulses on the interior of a uniform (i.e. gap-free) glass slab.

3.1. Explanation of the self-focusing phenomenon

When an ultrafast laser is focused inside a glass slab, the laser-irradiated area forms a filamentous structure due to the phenomenon of self-focusing by nonlinear absorption of an ultra-fast laser pulse.

In the transparent region of many materials, there is a nonlinear relationship between refractive index and propagating light field:

$$n = n_0 + n_2 |E(t)|^2 = n_0 + \tilde{n}_2 I(t) \quad (5)$$

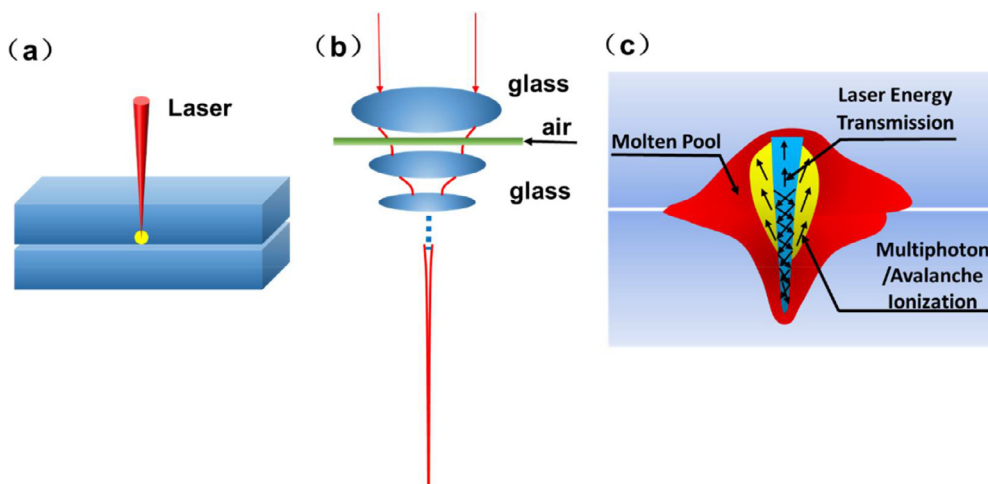


Fig. 5. Glass welding by ultra-fast laser (a) Two pieces of glass is illuminated by ultra-fast laser (b) An air layer is added to the self-focusing effect (c) Hypothesis of welding process.

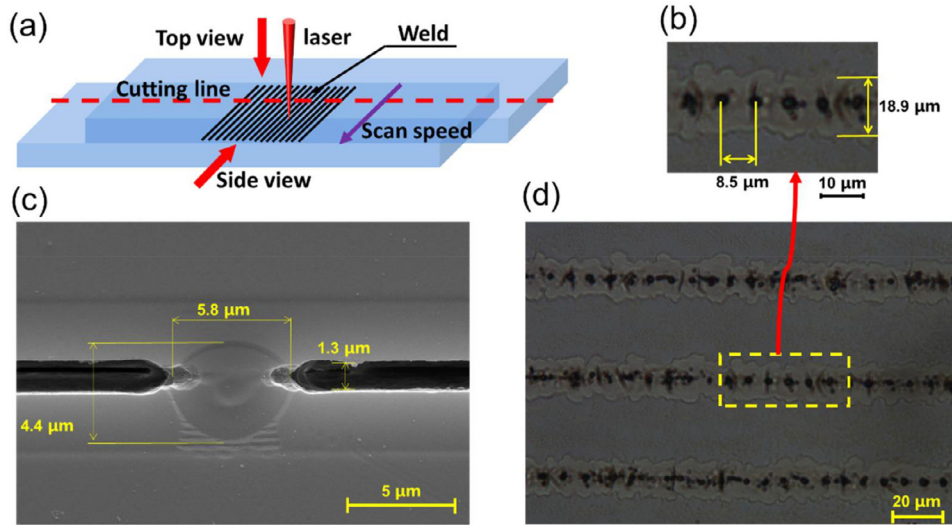


Fig. 6. Morphology of welding area (a) Schematic of sample cutting and observation (b) The larger version (top view) (c) The cross section of welding area observed by SEM (side view) (d) Top view.

where n_2 is the nonlinear refractive index coefficient, which represents the contribution to the refractive index of the nonlinear optical field. For femtosecond pulses, this nonlinearity is mainly the nonresonant nonlinear contribution of noninertial electron sources. Self-focusing is the induced lensing effect, which is caused by the distortion of the laser wavefront as it passes through the induced nonlinear medium. When a laser beam with Gaussian transverse distribution propagates in the medium, the refractive index of the medium is

$$n = n_0 + \Delta n(|E|^2) \quad (6)$$

where $\Delta n(|E|^2)$ is the refractive index change caused by the strong light.

If Δn is positive, the change of refractive index of the center part is larger than that of the edge part of the beam because of the greater light intensity in the central part of the beam, so the beam travels more slowly at the center than at the edges. As a result, the wave surface of the laser beam propagating through the medium becomes increasingly distorted, as shown in Fig. 3(b). The wavefront distortion is equivalent to that of a beam passing through a positive lens, and the light itself appears to be self-focused.

The cause of self-focusing of light beam is the change of photorefractive index. When the response time of refractive index change is on the order of fs, the distribution of electrons in the molecules or atoms is changed by the strong laser field. This leads to the change of the macroscopic electric polarization of the medium, which changes the refractive index. The high energy density laser beam focuses within the glass, as shown in Fig. 3(a). The refractive index of the glass changes in response to the light intensity, as shown in Fig. 3(b). The transverse distribution of the refractive index of the irradiated material is bell-shaped, when the laser with Gaussian distribution beam is powerful enough to produce nonlinear absorption effect. As a result, the irradiated material acts as a positive lens to focus the laser beam. The irradiated area in the glass is presented as a thin filamentous structure with a broad head, as shown in Fig. 3(c).

3.1.1. Experimental observations

We studied the effect of an air gap within the glass by initially directing the ps laser beam into a uniform glass sample, without an air gap. The ultra-fast laser with a repetition frequency of 100 kHz was focused inside a uniform glass plate at a scanning speed of 1000 mm/s.

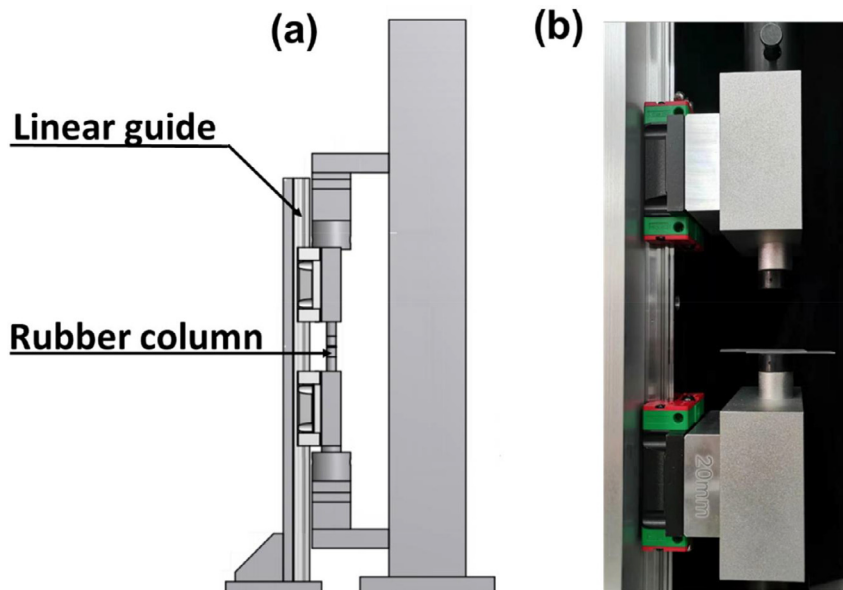


Fig. 7. (a) Schematic of longitudinal tensile test b) Tensile test setup.

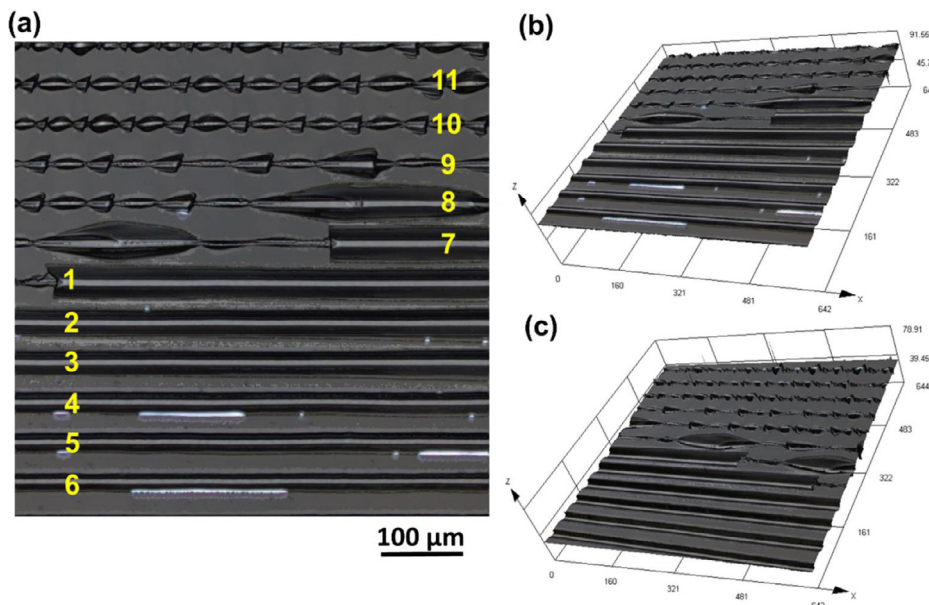


Fig. 8. Fracture morphology of the welded area observed by confocal microscopy (a) Morphology of bottom surface of the upper glass (b) Three-dimensional view of bottom surface of the upper glass. (c) Three-dimensional view of the upper surface of the bottom glass.

The spacing between successive irradiated regions was $10\text{ }\mu\text{m}$. The incident laser spot diameter was about $12\text{ }\mu\text{m}$, and the diameter of laser-irradiated area inside the glass was about $6\text{ }\mu\text{m}$. We observed that the pulse points are evenly distributed with spacing of $10\text{ }\mu\text{m}$ in Fig. 4(a). Fig. 4(b) shows an expanded detail of the same image. Each irradiated spot shows two distinct regions, namely an inner small circle with a diameter of $1.89\text{ }\mu\text{m}$ and an outer circle of diameter $5.16\text{ }\mu\text{m}$. We observed the cross-section of the laser-irradiated area, as shown in Fig. 4(c). The laser-irradiated area shows a filament-like structure consistent with the proposed structure in Fig. 3(c)

3.2. Ultra-fast laser welding of two glass plates

Next, we studied the welding of two identical glass plates, which we model as a process in which a thin layer of air with a refractive index of n_0 is inserted into a medium with a refractive index of n_1 , as shown in Fig. 5(b). The air gap and position of the focus will affect the self-focusing and thereby the weld. The effect of a gap between two glass surfaces and focal position of laser on welding is modelled by varying the thickness and position of air layer. As shown in Fig. 5(c), when the ultra-fast laser focuses on the interface between two glass samples, the high-energy laser induces multiphoton ionization in glass, and speed of the ionizing free electrons increases by absorbing laser photon energy. Then the electrons collide with other atoms, producing avalanche ionization. The coupling of electrons and phonons increases the temperature of ions and unionized glass, which rises to the melting temperature of glass, resulting in a successful weld. There is molten pool behavior flowing in the air gap [25].

We studied the change in weld morphology brought about by the presence of the air gap by comparing laser-irradiated uniform glass (Fig. 4) with the welding of two glass plates in close proximity but not in optical contact (Fig. 6). Fig. 6(a) shows the top view of welded area, and the viewing angles are shown in Fig. 6(d). Compared with Fig. 4, the area irradiated by a single laser pulse forms a welding beam with a width of $18\text{ }\mu\text{m}$ due to welding pool behavior. The shape of the irradiated and fused glass volume varies considerably from pulse to pulse. Fig. 6(c) is the larger version of red box area of Fig. 6(a). The single point spacing is about $8.5\text{ }\mu\text{m}$, which is smaller than the spacing in Fig. 4, even though we used the same processing parameters. Therefore, the presence of the air layer must have a great impact on the interaction between the laser light and the glass.

Fig. 6(d) shows the cross section of weld area observed by SEM after cutting, polishing and etching with hydrofluoric acid. The viewing directions are shown in Fig. 6(a). The welding morphology was still satisfactory when the gap between glass surfaces was $1.35\text{ }\mu\text{m}$, showing a tear-drop structure consistent with the hypothesis in Fig. 5(c). The whole weld area is relatively flat with very good bonding in the central area and signs of hydrofluoric acid etching on the left and right sides. There are slight pits in the central area and folds in the lower part of the tear-drop structure. We will conduct in-depth research into the defects in the welded area in future work.

The plasma density near the focus of the laser beam is higher than the critical density. That plasma will become denser as the laser energy increases. The focusing effect of laser beam is affected by photon-plasma scattering, which slightly attenuates the self-focusing and thereby distorts the original beam distribution. The self-focusing phenomenon in nonlinear absorption is affected by the thickness of air layer between two glass surfaces, thereby influencing the ionization strength of the gas. Therefore, the laser-irradiated regions appear irregular in shape.

3.3. Tensile test

If this laser welding technology is applied to the packaging of electronic components, the tensile strength perpendicular to the welded area is very important, because it directly affects the service performance. Therefore, we designed a longitudinal tensile test, as shown in Fig. 7. A linear guide rail was employed to ensure that the chuck motion was perpendicular to the surface of the device under test and a uniform force was applied. The welded sample was bonded to the surface of a detachable rubber ring with epoxy adhesive, as shown in Fig. 7(b). The device under test was mounted with the laser-irradiated surface uppermost. During test, the lower chuck remains fixed and the upper chuck moves upward at the speed of 0.5 mm/s . The fracture morphology of the welding area is shown in Fig. 8.

As shown in Fig. 8(a), the morphology in different welded regions is different with the same process parameters, because differences in the air gap thickness affect the absorption rate of the laser beam and the laser focal position. The air gaps in welded areas labeled 1 to 6 becomes smaller gradually and the gap at No. 6 welding line is minimum. While the areas labeled 7 to 11 becomes larger gradually and the gap at No. 11 welding line is maximum. The welded areas labeled 1 to 6 form relatively uniform straight lines, and were created where the air gap be-

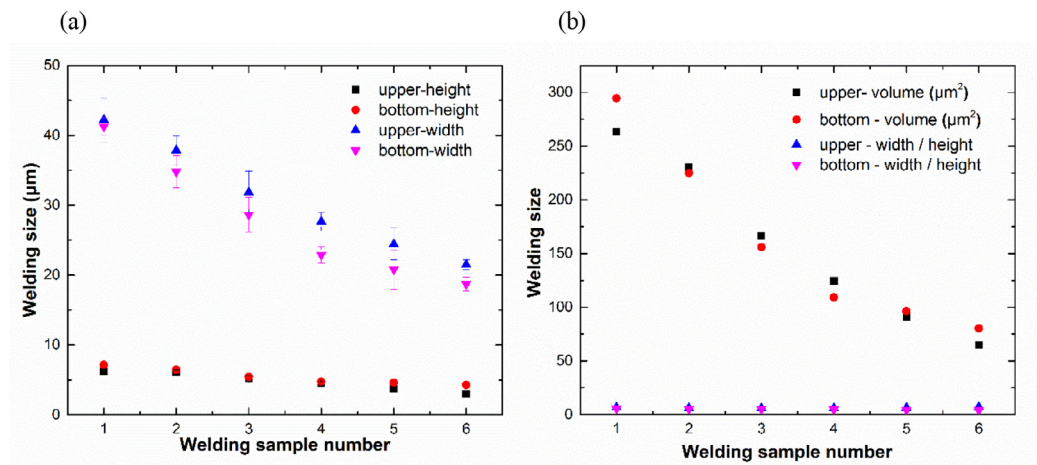


Fig. 9. Weld size at different areas (The height, width and volume refer to the height, width and volume of the molten zone respectively.).

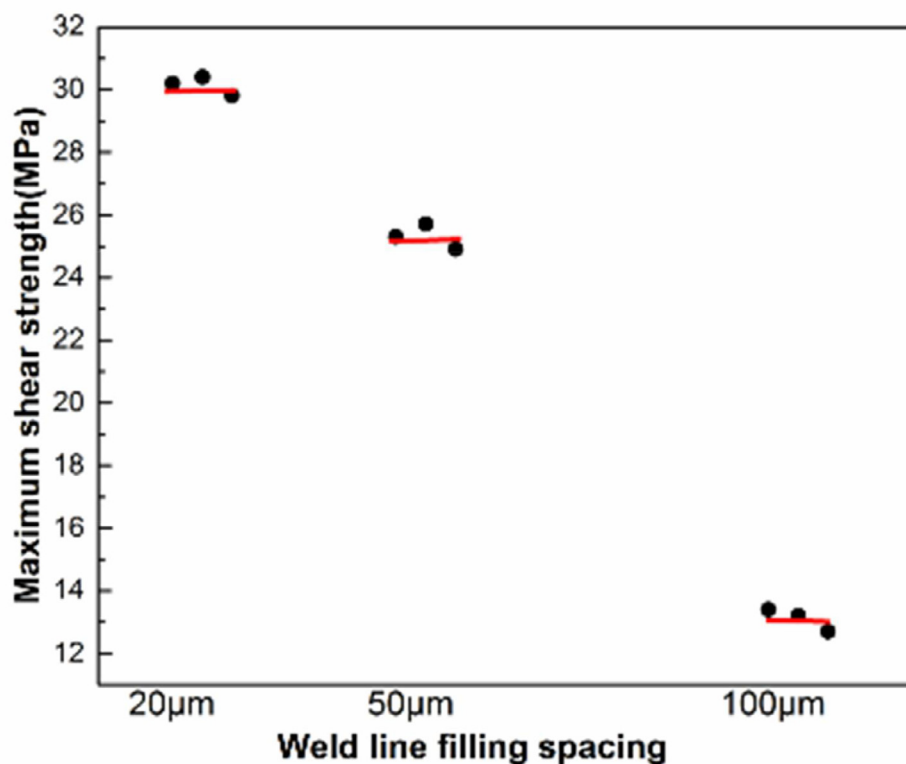


Fig. 10. Tensile strength test of different filling spacing. The points represent individual tests, and the line is the average value. (Weld line filling spacing refers to the distance between each welding line.).

tween glass plates is small. For regions 7 to 11, with the increase of air gap thickness, the weld fracture morphology does not form continuous straight lines, but rather rows of scattered small areas with a somewhat regular pattern.

Figs. 8(b) and (c) are respectively three-dimensional fracture morphology of the facing surfaces of the upper and bottom welded glass. We see that the welded area of upper glass presents a convex structure when viewed from below, while the bottom glass presents a concave structure when viewed from above, indicating that the bonding of the two glass samples is very good. This is also consistent with Fig. 6(b). The laser irradiated areas form a reliable fusion zone, with a longitudinal tension of up to 196 N between the two pieces of welded glass. The tensile strength was calculated by the ratio of the tested tension value to the welded area, up to 30 MPa. The weld sizes for weld lines 1 to 6 are shown in Fig. 9. We define the width of the welding area in the sample

after tension test as the width, and the convex part of the welding area in Fig. 8(b) and the concave part of the welding area in Fig. 8(c) are defined as height.

As the air gap of welded areas labeled 1 to 6 increases, the width and height of 1 to 6 weld lines decrease, and the variation trend of width is much larger, as shown in Fig. 9(a). The effective melt flow rate of glass after laser excitation is constant with the same processing parameters. When the air gap is small, both the width and height of the effective bonding zone are larger in the process of rapid melting and solidification, as shown in Fig. 5(c). As the air gap increases, the effective weld volume decreases, and the width-to-height ratio of each welding line is basically constant, as shown in Fig. 9(b). We used the tensile test to assess the quality of welded assemblies. In addition to the variation of filling spacing, the number of welding lines and laser processing parameters of each group of samples are consistent. Our

Table 1

Nonlinear index parameter of glass [26].

Nonlinear Source	Material	$\bar{n}_2/(cm^2/W)$	Response Time/s
Electron Nonresonance	Glass	10^{-16} — 10^{-15}	10^{-15} — 10^{-14}

tests show how the weld strength varies with weld line filling spacing (Fig. 10). When the two glasses are welded together, the bonding area has a strong adsorption force making the gap around welding area becomes smaller. Multiple welds will strengthen the weld when the filling spacing is small within a certain spacing interval. As shown in Fig. 10, when the filling spacing of the welding line is 20 μm , the tensile strength is about 30 MPa at most, and about 13 MPa when the filling spacing is 100 μm .

4. Conclusion

We obtained welded samples with good light transmission and sealing performance and a vertical strength up to 30 MPa using a 1064 nm ultrafast laser with 2 ps pulsewidth and average output power of 50 W. The cross section of the welded area is a complete tear-drop structure observed by SEM. The thickness of the air layer can be determined from measurement of interference fringes, and the width and depth of the effective melt zone decreased with the increase of air gap, but the ratio of height to width remains constant. When the air gap is 2 μm , high strength welds with smooth microstructure can be achieved. This is of great significance for the packaging of electronic components. Self-focusing of the incident laser beam caused by ultrafast laser pulses in glass is affected by the additional air layer between the samples to be welded, which can result in variations from one weld spot to another. The effective width and depth of the tear-drop structure caused by a multi-photon effect in the welding regions vary with the thickness and position of air layer. Considering the thickness and position of the air gap as the constraint conditions can help us explore the law of laser transmission and absorption in the microscopic region of the weld, which is conducive to the industrial application of this welding technology. The vertical tensile strength of the welds decreases with the increase of filling spacing of welding lines. In the future work, we will test the sealing fatigue residual stress in combination with practical applications such as air capacitor encapsulation, and conduct basic research on the process of formation process of welded areas in order to explore the ps-laser welding mechanism.

Table 1

Author statement

Jiejuan Zhang: Writing, Methodology, Conceptualization, Implementation, Sample preparation

Shanghang Cheng: Welding samples, Data curation; **Hanlun Lu:** Polishing the samples;

Minghe Huang: Tensile test, Morphology observation; **Jiaming Li:** Analyzing the data;

Liang Guo: Supervision; **Qitao Lue:** Ultra-fast laser equipment manufacturing and operation; **Qingmao Zhang:** Reviewing and Editing, Experiment leader

Declaration of Competing Interest

There is no conflict of interest in this work.

Acknowledgments

This work was financially supported by the National Key Research and Development Program of China (2017YFB1104500); Department of Science and Technology of Guangdong Province (2018B030323017); Key-Area Research and Development Program of Guangdong Province (2020B090922006); Young Innovative Talents Project in Universities of Guangdong Province (2018KQNCX057); Young Scholar Foundation of South China Normal University (19KJ13). Featured Innovation Project of Guangdong Education Department (2019KTSCX034)

References

- [1] Zhang G, Bai J, Zhao W, Zhou K, Cheng G. Interface modification based ultrashort laser microwelding between SiC and fused silica. *Opt Express* 2017;25:1702–9.
- [2] Tan A, Tay F. Localized laser assisted eutectic bonding of quartz and silicon by Nd: YAG pulsed-laser. *Sensors and Actuators A: Physical* 2005;120:550–61.
- [3] Pohl L, von Witzendorff P, Chatzizyrl E, Suttman O, Overmeyer L. CO2 laser welding of glass: numerical simulation and experimental study. *The International Journal of Advanced Manufacturing Technology* 2017;90:397–403.
- [4] Cozma A, Puers B. Characterization of the electrostatic bonding of silicon and Pyrex glass. *Journal of Micromechanics and Microengineering* 1995;5:98.
- [5] Horn A, Mingareev I, Werth A, Kachel M, Brenk U. Investigations on ultrafast welding of glass–glass and glass–silicon. *Applied Physics A* 2008;93:171.
- [6] Hélie D, Lacroix F, Vallée R. Reinforcing a direct bond between optical materials by filamentation based femtosecond laser welding. *Journal of Laser Micro Nanoengineering* 2012;7:284.
- [7] Tamaki T, Watanabe W, Nishii J, Itoh K. Welding of transparent materials using femtosecond laser pulses. *Jpn J Appl Phys* 2005;44:L687.
- [8] Cvecek K, Miyamoto I, Strauss J, Bui V, Scharfenberg S, Frick T, Schmidt M. Strength of Joining Seams in Glass Welded by Ultra-fast Lasers Depending on Focus Height. *Journal of Laser Micro Nanoengineering* 2012;7:68–72.
- [9] Richter S, Doering S, Tünnemann A, Nolte S. Bonding of glass with femtosecond laser pulses at high repetition rates. *Applied Physics a-Materials Science & Processing* 2011;103:257–61.
- [10] Sugioka K, Iida M, Takai H, Micorikawa K. Efficient microwelding of glass substrates by ultrafast laser irradiation using a double-pulse train. *Opt Lett* 2011;36:2734–6.
- [11] Gattass RR, Mazur E. Femtosecond laser micromachining in transparent materials. *Nat Photonics* 2008;2:219.
- [12] Sundaram S, Mazur E. Inducing and probing non-thermal transitions in semiconductors using femtosecond laser pulses. *Nat Mater* 2002;1:217.
- [13] Wu S, Wu D, Xu J, Wang H, Makimura T, Sugioka K, Midorikawa K. Absorption mechanism of the second pulse in double-pulse femtosecond laser glass microwelding. *Opt Express* 2013;21:24049–59.
- [14] Miyamoto I, Horn A, Gottmann J. Local melting of glass material and its application to direct fusion welding by ps-laser pulses. *J. Laser Micro/Nanoeng* 2007;2:7–14.
- [15] Richter S, Döring S, Tünnemann A, Nolte S. Bonding of glass with femtosecond laser pulses at high repetition rates. *Applied Physics A* 2011;103:257–61.
- [16] Miyamoto I, Okamoto Y, Tanabe R, Ito Y, Cvecek K, Schmidt M. Mechanism of dynamic plasma motion in internal modification of glass by fs-laser pulses at high pulse repetition rate. *Opt Express* 2016;24:25718–31.
- [17] Roth G-L, Rung S, Hellmann R. Welding of transparent polymers using femtosecond laser. *Applied Physics A* 2016;122:86.
- [18] Richter S, Zimmermann F, Sutter D, Budnicki A, Tünnemann A, Nolte S. Ultrashort pulse laser welding of glasses without optical contacting. *Proceedings of the SPIE* 2017;10094:1009418 10094111009411 (1009418 pp.).
- [19] Chen J, Carter RM, Thomson RR, Hand DP. Avoiding the requirement for pre-existing optical contact during picosecond laser glass-to-glass welding (vol 23, pg 18645, 2015). *Opt Express* 2015;23:28104–5.
- [20] Cvecek K, Miyamoto I, Strauss J, Wolf M, Frick T, Schmidt M. Sample preparation method for glass welding by ultrashort laser pulses yields higher seam strength. *Appl. Opt* 2011;50:1941–4.
- [21] Cvecek K, Odato R, Dehmel S, Miyamoto I, Schmidt M. Gap bridging in joining of glass using ultra short laser pulses. *Opt Express* 2015;23:5681–93.
- [22] Gstalter M, Chabrol G, Bahouka A, Dorkenoo K-D, Rehspringer J-L, Lecler S. Long focal length high repetition rate femtosecond laser glass welding. *Appl. Opt.* 2019;58:8858–64.
- [23] Cvecek K, Dehmel S, Miyamoto I, Schmidt M. A review on glass welding by ultra-short laser pulses. *International Journal of Extreme Manufacturing* 2019;1.
- [24] Cvecek K, Miyamoto I, Schmidt M. Gas bubble formation in fused silica generated by ultra-short laser pulses. *Opt Express* 2014;22:15877–93.
- [25] Zhang J, Xu S, Dong Y, Zhang C, Li J, Guo L, Zhang Q. Microwelding of glass to silicon by green ultrafast laser pulses. *Opt Laser Technol* 2019;120.
- [26] Fournier J, Snitzer E. The nonlinear refractive index of glass. *IEEE J Quantum Electron* 1974;10:473–5.